SIMULATED
Asset Management

Performance decomposition

Agent: **Portfolio Management**

OPENING PROMPT

```
Break down the performance attribution for SAM Global Flagship portfolio versus S&P 500 for
the most recent quarter. What drove our performance?
```

The demo date is the latest price date in the Snowflake Public Data share at build time, and positions are month-end. Agents answer from live data, so figures, names and rankings in your environment can differ from the examples in this script.

Business Context Setup

Persona: Victoria Chen, Chief Investment Officer at Simulated Asset Management
Business Challenge: Understanding the true drivers of portfolio performance requires decomposing returns into allocation decisions vs. security selection vs. factor exposures. Traditional attribution reports are static, backward-looking, and require multiple systems. The CIO needs instant answers to board questions about why portfolios outperformed or underperformed benchmarks, what hidden risks exist, and how portfolios would behave under stress scenarios.
Value Proposition: AI-powered attribution intelligence provides instant, multi-dimensional performance decomposition combining Brinson attribution, factor analysis, hidden risk detection, and stress testing in a single conversational interface. Real-time answers enable confident board presentations and rapid investment decision-making.

Agent: `AM_portfolio_management_copilot`
Data Available: 24 months of monthly attribution data across 11 portfolios, 11 GICS sectors, 7 systematic factors, 5 hidden factors, 10 stress scenarios. Brinson and factor attribution are computed from **real market data** (stock prices, sector ETF returns, and SEC financials sourced via Snowflake Marketplace). Portfolio weights are dynamic based on real close prices.

Demo Flow

Scene Setting: Victoria is preparing for the quarterly board meeting where she needs to explain portfolio performance relative to its benchmark. The board will want a clear decomposition of returns — whether the portfolio outperformed or underperformed — and will ask tough questions about risk exposures and how the portfolio would perform in a crisis.

Note: Attribution data reflects real market conditions. In the current build the latest quarter is an **underperformance** quarter for SAM Global Flagship Multi-Asset, and the expected responses below describe that case. **Before presenting**, run the Step 1 question to confirm. Both outcomes make compelling demos:

- **Outperformance quarter**: Highlights what's working — which sector bets and stock picks are generating alpha
- **Underperformance quarter**: Often the *stronger* demo — shows the agent diagnosing problems, identifying the exact source of the drag, and recommending corrective actions. This is when a CIO needs the tool most.

Step 1: Brinson Attribution Overview

Skill Activated: `multi-level-attribution` **Workflow**: Sector attribution -> STOPPING POINT -> [user choice] -> drill-down -> synthesis **Demo Value**: One question triggers a structured analytical workflow with interactive checkpoints

User Input:

> **PROMPT**
>
> `Break down the performance attribution for SAM Global Flagship portfolio versus S&P 500 for the most recent quarter. What drove our performance?`

Tools Used:

- `server_skill` — Loads multi-level-attribution skill
- `attribution_analyzer` (Cortex Analyst) - Retrieves Brinson decomposition into allocation, selection, and interaction effects

Expected Response:

- **Attribution Summary**: Flagship finished **essentially in line with, marginally behind** the S&P 500 over the latest quarter (both up; active return a few basis points negative)
- **Allocation-driven**: sector allocation and interaction were both heavy drags, while stock selection was **net positive**
- **Sector Breakdown**: **Industrials** was the largest detractor and **Consumer Staples** the second (a pure allocation drag); **Information Technology** was the best contributor, almost entirely through selection
- **Key Insight**: this was a top-down positioning problem, not a stock-picking problem
- **Brinson Identity Check**: Active return = Allocation + Selection + Interaction; the linked effects reconcile to the active return within about 1 bp
- **Actionable Recommendations**: review the Industrials and Consumer Staples positioning; protect the selection engine in IT
- **Stopping point**: drill-down options by country, industry or linked periods

Talking Points:

- Brinson-Fachler attribution is the industry standard for performance decomposition
- Instant decomposition eliminates days of manual analysis
- Clear separation of "right sectors" (allocation) vs "right stocks" (selection) - here the stocks were right and the sectors were wrong
- The agent adapts its narrative to the data — it doesn't assume good or bad performance, it diagnoses what actually happened and recommends next steps

Key Features Highlighted:

- Cortex Analyst semantic layer over attribution fact tables
- **Computed from real stock prices (sourced via Snowflake Marketplace)** — portfolio sector returns are weighted averages of actual stock returns, not synthetic data
- Benchmark returns from real SPDR sector ETFs (XLK, XLF, XLV, etc.)
- Industry-standard Brinson-Fachler methodology
- Adaptive narrative: works equally well for outperformance and underperformance quarters

Step 1b: Multi-Level Attribution Drill-Down (Stopping Point Continuation)

Presenter Transition:

> *"Notice the agent paused and offered drill-down options — this is the multi-level-attribution skill's stopping point. The PM can choose to drill into a sector, pivot to country, or see linked periods. This is the key differentiator from traditional attribution tools — it's not just data retrieval, it's a structured analytical workflow with checkpoints. Let me choose country..."*

Reasoning: The skill's stopping point demonstrates interactive workflow orchestration. The agent doesn't just dump data — it presents a summary, then offers contextual next steps.

User Input:

> **PROMPT**
>
> `Now show me this by country — which countries drove outperformance last quarter?`

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Queries FACT_BRINSON_ATTRIBUTION_DETAIL with grouping_dimension = 'COUNTRY'

Expected Response:

- The book resolves to three buckets: **US**, **Taiwan** and an **Unclassified** cross-border/multi-asset bucket
- **Unclassified** is the only positive contributor (allocation-driven); **US** is the largest detractor through selection; **Taiwan** is an allocation drag
- Country effects sum to the same active return as the sector view
- The agent notes that US selection reads negative while sector selection was strongly positive - so the sector lens is the more informative one

Presenter Note: Sector, country and industry decompositions all sum to the same linked active return, so the totals tie across lenses; only the story told by each lens differs.

Talking Points:

- Same Brinson-Fachler methodology applied to any grouping dimension
- The Rainardi insight: "same data, different classification produces different attribution results" — sector vs country decomposition tells different stories
- Country attribution is particularly valuable for global mandates

Key Features Highlighted:

- Multi-level attribution from a single generalised table (FACT_BRINSON_ATTRIBUTION_DETAIL)
- Instant pivot between sector, country, and industry without recalculation
- Industry drill-down via parent_grouping_value hierarchy

User Input (follow-up):

> **PROMPT**
>
> `Drill down into Information Technology — what industries within tech contributed?`

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Queries grouping_dimension = 'INDUSTRY' with parent_grouping_value = 'Information Technology'

Expected Response:

- Industry breakdown for the same quarter, using a granular (SIC-style) industry taxonomy mapped to the tech complex
- **Semiconductors** contributed positively through genuine stock selection
- **Prepackaged software** was the largest tech drag, losing on both allocation and selection; data-processing services also detracted
- **Computer integrated systems design** is the largest single contributor, almost entirely allocation-led
- Conclusion: the IT result is carried by systems-design and semiconductor names; software is the soft spot

Talking Points:

- Hierarchical drill-down from sector to industry — the Rainardi pattern from real AM data warehouses
- This is how portfolio managers actually think: "Tech overweight worked, but was it semis or software?"
- In production, this hierarchy would extend to individual securities

Step 2: Sector and Holding-Level Drill-Down

Presenter Transition:

> *"The investment committee will want to know exactly which sectors moved the needle, and which stocks inside them. Let me look at the latest month on its own, then at the holdings..."*

Reasoning: Sector attribution identifies contributors; position-level analysis provides accountability.

User Input:

> **PROMPT**
>
> `Which sectors had the biggest impact on our active return last month? Show me both allocation and selection effects by sector.`

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Query FACT_BRINSON_BY_SECTOR for detailed sector breakdown

Expected Response:

- Latest month (September) is a different picture from the quarter and is again allocation-driven
- **Health Care** is the largest detractor (allocation, plus weak selection)
- **Consumer Staples**, **Industrials** and **Communication Services** are the top contributors - Industrials has flipped from the quarter's worst sector to a positive
- IT allocation hurt but its selection was a strong offset; Communication Services leads on selection
- The agent notes the quarter's damage was front-loaded earlier in the period

Presenter Transition:

> *"Stock selection in tech keeps coming through. Let me see which holdings did the work..."*

User Input (follow-up):

> **PROMPT**
>
> `Which technology stocks drove our selection effect? What was our allocation to healthcare vs. the benchmark?`

Tools Used:

- `portfolio_analyzer` (Cortex Analyst) / SQL - Holding active weights and quarter-to-date returns

Expected Response:

- The agent explains that attribution resolves to sector and industry, **not to individual positions**, and instead pairs each IT holding's active weight with its holding return
- Selection was carried by overweights that worked: **AMD** (the dominant IT position) and **Microsoft**, with a smaller contribution from Adobe
- Offsets: being underweight **Snowflake** and **Salesforce** during their rallies cost performance; underweights in IBM and Intel helped
- Health Care: a **large structural underweight** versus the benchmark

Talking Points:

- Sector-level granularity for precise attribution
- Holding-level accountability through active weight and return, with the agent stating plainly what the attribution model does and does not store
- Multi-level hierarchy: Portfolio, Sector, Industry, Holding

Key Features Highlighted:

- Sector-level Brinson decomposition with contribution ranking
- Holding-level active weights and returns
- Sector weight comparison

Step 3: Factor Attribution Drill-Down

Presenter Transition:

> *"The Brinson analysis says sector positioning cost us and stock picking helped. Board members will ask whether factor bets explain any of this. Let me decompose the factor exposures..."*

Reasoning: Brinson shows sector effects but doesn't explain systematic risk exposures. Factor attribution provides the "why behind the why" for sophisticated board members.

User Input:

> **PROMPT**
>
> ```
> What factor exposures drove our returns? Show me the contribution from all systematic
> factors.
> ```

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Retrieves factor-based return attribution

Expected Response:

- **Headline**: factor contributions are **small** and net slightly negative - **Momentum** is the largest drag, Market and Value small drags, Size, Quality and Growth small positives
- **Size** carries the largest loading but contributed only a few basis points because the size premium was near zero
- **Conclusion**: this was a sector-specific quarter, not a factor-driven one
- **Factor Contributions**: Return contribution from each systematic factor, computed from **real data**:

Factor	How Computed
Market	Real beta from regression of stock returns vs SPX
Value	Real P/E ratios from SEC financial filings
Growth	Real revenue growth from SEC filings
Momentum	Real 12-month price momentum
Quality	Real ROE + operating margin + leverage from SEC filings
Size	Real market cap (price x shares outstanding)
Volatility	Real 60-day rolling standard deviation of returns

- **Exposure**: Portfolio-weighted average of security-level factor scores from FACT_FACTOR_EXPOSURES
- **Factor Returns**: Long-short quintile returns (top vs bottom 20% of stocks by factor score)

Talking Points:

- Factor attribution computed from **real security-level data** — not synthetic
- Exposures from real regression betas, SEC financials, and market cap data
- Factor returns from actual long-short quintile portfolio performance
- Identifies whether returns came from skill or factor bets — critical for both good and bad quarters

Key Features Highlighted:

- Real factor model built from stock prices and SEC financial data (sourced via Snowflake Marketplace)
- Portfolio-weighted exposure aggregation from 503-security universe
- Quintile-based factor return estimation (industry-standard methodology)

Step 4: Hidden Factor Detection

Presenter Transition:

> *"Systematic factors barely moved the needle. But could there be hidden concentrations - AI exposure, geopolitical risk - that traditional models aren't capturing? This is where AI goes beyond standard attribution..."*

Reasoning: Hidden factors represent uncompensated or emerging risks not captured by traditional factor models. This demonstrates AI sophistication beyond standard attribution.

User Input:

> **PROMPT**
>
> ```
> What hidden or alternative factor exposures exist in the portfolio? Are we unknowingly
> concentrated in AI-related stocks or exposed to geopolitical risks?
> ```

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Retrieves AI-detected hidden factor exposures

Expected Response:

- **Hidden Factor Exposures**: Portfolio exposure to non-traditional factors

Factor	Description
AI_Exposure	Concentration in AI/ML beneficiary stocks
Reshoring_Benefit	Supply chain relocalisation exposure
Rate_Convexity	Interest rate sensitivity beyond duration
Climate_Transition	Green transition winners/losers exposure
Geopolitical_Risk	Country/region tension sensitivity

- **Findings**:
  - **AI_Exposure** is by far the largest hidden tilt - the hidden version of the AMD, Microsoft and semiconductor concentration
  - **Climate_Transition** is a sizeable second tilt
  - **Reshoring_Benefit** is moderate but has the highest explanatory power
  - **Geopolitical_Risk** is a mild positive (TSM the obvious contributor)
  - **Rate_Convexity** is slightly negative - little hidden rate sensitivity
- **Answer to the question**: yes on AI, only mildly on geopolitics

Talking Points:

- Hidden factors are computed from **real SEC filing data and AI analysis of earnings call transcripts** — not simulated
- AI_Exposure combines keyword matching on SEC business segments (Cloud, Data Center, GPU, etc.) with AI_AGG scoring of earnings call text (uses AI because transcript text is unstructured)
- Geopolitical_Risk is a **deterministic SQL calculation** — geographic revenue from SEC segments classified via a risk-tier lookup table (DIM_GEO_RISK_CLASSIFICATION: 189 geographies -> HIGH/MEDIUM/LOW), weighted by revenue share with concentration bonuses. No AI needed for structured numeric data.
- Rate_Convexity uses real debt-to-equity and current liability ratios from SEC financials
- Climate_Transition maps GICS sector carbon intensity combined with ESG environmental scores
- All factors are z-scored cross-sectionally and aggregated using real portfolio position weights
- In production, these could also be sourced from specialist providers such as MSCI (Barra thematic factors), Axioma (custom risk models), BlackRock Aladdin (thematic baskets)

Key Features Highlighted:

- AI-powered factor discovery
- Thematic risk detection
- Forward-looking risk identification

Step 5: Stress Scenario Analysis

Presenter Transition:

> *"The board will inevitably ask 'what happens if there's a market crash?' Let me show you how we can instantly stress test the portfolio against historical and hypothetical scenarios..."*

Reasoning: Board members always want to understand downside scenarios. Stress testing demonstrates robust risk management and preparedness.

User Input:

PROMPT

```
How would our portfolio perform in a 2008-style financial crisis? Also show me the impact of a sudden 200bp rate hike scenario.
```

Tools Used:

- `backtest_historical_stress` (Stored Procedure) - Replays the 2008 financial crisis against current holdings

- `scenario_sensitivity` (Stored Procedure) - RATE_SHOCK scenario for the 200bp rate hike

Expected Response:

- **2008-style replay**: a painful absolute loss but **much milder than the market** - the multi-asset book cushions a systemic crash
- **+200bp rate hike**: a modest loss, roughly double the +100bp case (broadly linear in shock size)
- In both, the loss is driven almost entirely by **Market beta**; low-volatility and size tilts provide a small cushion
- The slightly negative Rate_Convexity from Step 4 means there is no hidden leverage amplifying the rate shock
- **Mitigation Options**: the agent discusses reducing equity beta rather than rate hedges

Talking Points:

- Instant stress testing enables confident board responses
- Both historical and hypothetical scenarios available
- Connects factor exposures to real-world crisis impacts

Key Features Highlighted:

- Pre-built historical and hypothetical stress scenarios
- Factor-based stress transmission
- Actionable risk mitigation suggestions

Step 6: Intelligent Driver Discovery (Synthesis)

Presenter Transition:

> *"We've looked at Brinson, factors, and hidden factors individually. But the real power is when the agent synthesises all three to tell us the true story of what drove performance..."*

Reasoning: Demonstrates cross-tool synthesis — the agent combines multiple data sources to deliver a unified narrative that goes beyond what any single analysis provides.

User Input:

> **PROMPT**
>
> ```
> Looking at everything together — Brinson, factor exposures, and hidden factors — what are
> the true drivers of our Flagship portfolio's performance? Is our alpha from real
> stock-picking skill, or are we riding factor tailwinds?
> ```

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Brinson, factor and hidden-factor attribution in one semantic view (multi-query synthesis)

Expected Response:

- **Unified Narrative**: The agent reconciles Brinson, factor and hidden-factor results in one table
- **True Alpha Assessment**: this quarter's positive selection is **genuine stock-picking**, because factors were a headwind and so cannot be its source - concentrated in AMD and semiconductors
- **The caveat**: the skill is **not persistent** - over the trailing 12 months alpha persistence is low and average selection is roughly flat, so this quarter is the exception rather than the rule
- **The leak**: allocation, not selection
- **Hidden risk**: the large AI tilt is thematic beta, not diversified selection

Talking Points:

- This is the "killer question" that boards actually ask — cross-tool synthesis required
- The agent combines three analytical perspectives (Brinson, factor, hidden factor) and synthesises the results intelligently
- Demonstrates that the platform isn't just retrieval — it's analytical reasoning

User Input (follow-up):

> **PROMPT**
>
> ```
> Based on that analysis, which specific decisions generated real alpha vs factor exposure?
> What should we do differently next quarter?
> ```

Expected Response:

- **Real alpha**: AMD and Microsoft overweights, plus smaller Adobe and Communication Services calls; at neutral sector weights the picks would have been worth a clear positive active return
- **Exposure, not skill**: the hidden AI tilt and market beta
- **Value-destroying decisions**: the **Industrials** weighting (the costliest by far), then **Consumer Staples**, with smaller Financials and Energy costs; the Health Care underweight is a standing risk; the IT overweight was mildly additive
- **Recommendations**: tighten the sector-allocation process first; keep the stock-selection process

Step 7 (Optional): Cross-Dimension Comparison & YTD Linked Attribution

Presenter Transition:

> *"Is this really a 'US tech' story or a broad technology story? The same excess return decomposed by sector and by country can tell very different stories. Let me compare both lenses, then extend to the full YTD view with linked attribution."*

Reasoning: Use for audiences with attribution specialists. It shows classification sensitivity and correct multi-period linking.

User Input:

> **PROMPT**
>
> ```
> Compare the sector attribution and country attribution for our Flagship portfolio last
> quarter. Which lens tells a more useful story?
> ```

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Queries both grouping_dimension = 'SECTOR' and 'COUNTRY'

Expected Response:

- Comparison table: the **sector lens wins** on granularity, reconciliation and actionability
- Sector view isolates the decisions (Industrials and Consumer Staples drags, IT offset); the country view collapses almost everything into a single US bucket
- The country view flips the sign on selection, the one thing that went right
- Only the sector view reconciles to the active return (see the presenter note in Step 1b)
- Bottom line: for a US-concentrated, S&P 500-benchmarked book, country attribution adds little

User Input (follow-up):

> **PROMPT**
>
> ```
> Show me the YTD linked attribution by sector for Flagship. How do the linked effects compare
> to the simple sum?
> ```

Tools Used:

- `attribution_analyzer` (Cortex Analyst) - Queries linked_attribution with period_type = 'YTD'

Expected Response:

- YTD linked attribution by sector using Frongello base-period adjustment - negative YTD, with **Consumer Staples** and **Industrials** the largest detractors
- The simple sum of monthly effects **overstates the YTD drag** slightly versus the linked figure
- Sector-level differences are small and sit mainly in the largest-magnitude sectors
- Explanation of why they differ (compounding), and a recommendation to always report the linked figure

Talking Points:

- This is the Rainardi insight: "same data, different classification produces different attribution results"
- In real AM, the classification choice matters — country of incorporation vs country of risk can tell different stories
- Frongello linking is forward-looking — future market movements don't affect past contributions
- No residuals — the linked effects sum exactly to the compounded active return
- This is the standard that performance teams use for multi-period client reporting

Key Features Highlighted:

- Multi-level attribution from a single FACT_BRINSON_ATTRIBUTION_DETAIL table
- Frongello base-period adjustment for multi-period linking
- Agent synthesises cross-dimension insights automatically

Step 8: Same Data, Three Audiences

Skill Activated: `attribution-report-generator` (detects "CIO-level" -> executive audience tier) **Workflow**: Sector attribution -> anomaly check -> synthesise into executive format **Demo Value**: The analysis is already done; the same data is re-presented for the board, the PM team and clients without re-running it

Presenter Transition:

> *"Victoria now has to explain the same quarter to three different audiences in the next hour: the board, the PM team and clients. Today that means three separate reports. Watch the agent change only the presentation layer..."*

User Input (board):

PROMPT

```
Generate a quarterly attribution report for SAM Global Flagship — board-level summary for the CIO.
```

Expected Response:

- The agent loads the attribution-report-generator skill
- Runs sector attribution + anomaly check automatically (Steps 1-3 of the skill)
- Presents executive-format output:
  - Headline: underperformed the benchmark; an allocation problem, not stock selection
  - Key drivers: Industrials and Consumer Staples allocation; positive selection led by IT; factors broadly neutral
  - Risk flags: **one MEDIUM item, no HIGH** (Health Care concentration and Market-factor dominance), plus the AI tilt and stress results
  - NO tables, NO individual positions
- Stops with: "Would you like the PM-level detail, a client version, or shall I add currency analysis?"

User Input (PM team):

PROMPT

```
Now give me the PM-level version with full attribution tables and factor breakdown.
```

Expected Response:

- Full attribution table with allocation, selection, interaction effects by sector
- Factor contribution breakdown with exposures
- Country and industry sections with the reconciliation caveat
- Decision accountability: the Industrials positioning was the wrong call; semiconductor selection was the right one
- Specific recommendations

User Input (client):

PROMPT

```
Now write this for a client report in plain English. No jargon, no factor names, no basis points.
```

Expected Response:

- "Your portfolio declined slightly this quarter while the market rose, leaving it behind the benchmark."
- Plain English explanations: the companies chosen did well (technology, a leading chip maker, Microsoft); the mix across areas of the market (more in industrials, very little in healthcare) held it back
- A note on positioning: a strong tilt towards technology and AI beneficiaries
- NO attribution terminology

Talking Points:

- Same data, three narratives — demonstrates adaptive AI communication
- The skill automatically detected "board-level" and applied executive formatting
- Eliminates the manual work of writing separate board, PM, and client reports
- Audience detection from question context keywords (CIO, board, client, plain English)

Scenario Wrap-up

Business Impact Summary:

- **Board Readiness**: Complete performance narrative prepared in minutes instead of days
- **Risk Transparency**: Hidden exposures and stress vulnerabilities identified proactively
- **Decision Speed**: Real-time attribution enables faster rebalancing decisions
- **Stakeholder Confidence**: Data-driven answers to any attribution question
- **One Analysis, Many Audiences**: Board, PM and client narratives from the same underlying data

Technical Differentiators:

- **Unified Attribution Platform**: Brinson, factor, hidden factor, and stress analysis in one agent
- **AI-Powered Hidden Factors**: Detection of emerging risks beyond traditional models
- **Real-Time Calculation**: No batch processing delays - instant answers from live data
- **Natural Language Interface**: No SQL or programming required for sophisticated analysis
- **Multi-Period Linking**: Frongello-linked QTD/YTD attribution suitable for formal client reporting